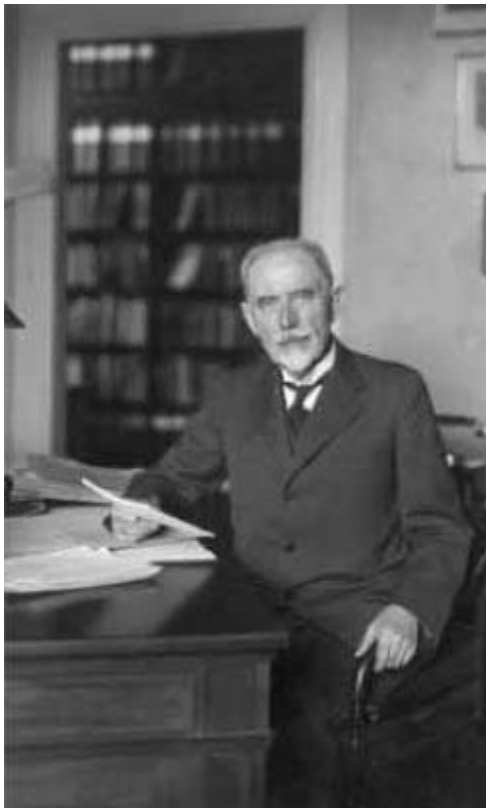
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41-52 minutes

2.1 Concept of pH



Søren Peder Lauritz Sørensen (1868-1939)

Born in Havrebjerg, Denmark, Sørensen was a Danish chemist, famous for the introduction of the concept of pH, a scale for measuring acidity and basicity. From 1901 to 1938 he was head of

the prestigious Carlsberg Laboratory, Copenhagen. While working at the Carlsberg Laboratory he studied the effect of ion concentration on proteins,and because the concentration of hydrogen ions was particularly important, he introduced the pH-scale as a simple way of expressing it in 1909.

The Danish scientist Sørensen defined the concept of pH as follows:

pH equals the inverse of the logarithm to the base 10 of the hydrogen ion concentration, as shown by the formula:

$$\mathbf{pH = -10log\ [H^+] = paH}^{1)}\ \mathbf{(1)}$$

Later Sørensen found this definition to be incorrect, since more concentrated solutions appeared to give deviations between calculated and measured values.

The definition therefore had to be modified to:

pH equals the inverse of the logarithm to the base 10 of the hydrogen ion **activity**[2)] as shown by the formula:

$$\mathbf{pH = -10log\ aH^+ = pH}^{3)}\ \mathbf{(2)}$$

The activity of the hydrogen ions is not always linear with the concentration, since this activity is not only affected by the concentration of ions, but also by other factors, such as:

- The activity of other ions present in the solution
- The temperature of the solution
- The character of the solution.

To facilitate the accurate measurement of pH, and its presentation as a scale, a range of "standard liquids" or "buffer solutions" are used.

These liquids, whose constituents are accurately defined, have known stable values.

Although in the preceding text the relationship to hydrogen ions has been made, research has shown, that the activity of hydroxonium ions (H_30^+) is more relevant. In aqueous solutions free H^+ ions do not occur, but are always in combination with water molecules.

$$H^+ + H_20 \leftrightarrow H_30^+$$

Consequently, a more correct definition for pH is:

$$pH = -10log\ aH_30^+ \quad (3)$$

For clarity, the notation H^+ will be used in the book as the hydroxonium ion.

Note 1. The notation -10log .... can also be written p ....
Note 2. See Appendix 2: Definitions.
Note 3. See Chapter 2.8: Buffer solutions.

2.2 The pH Scale

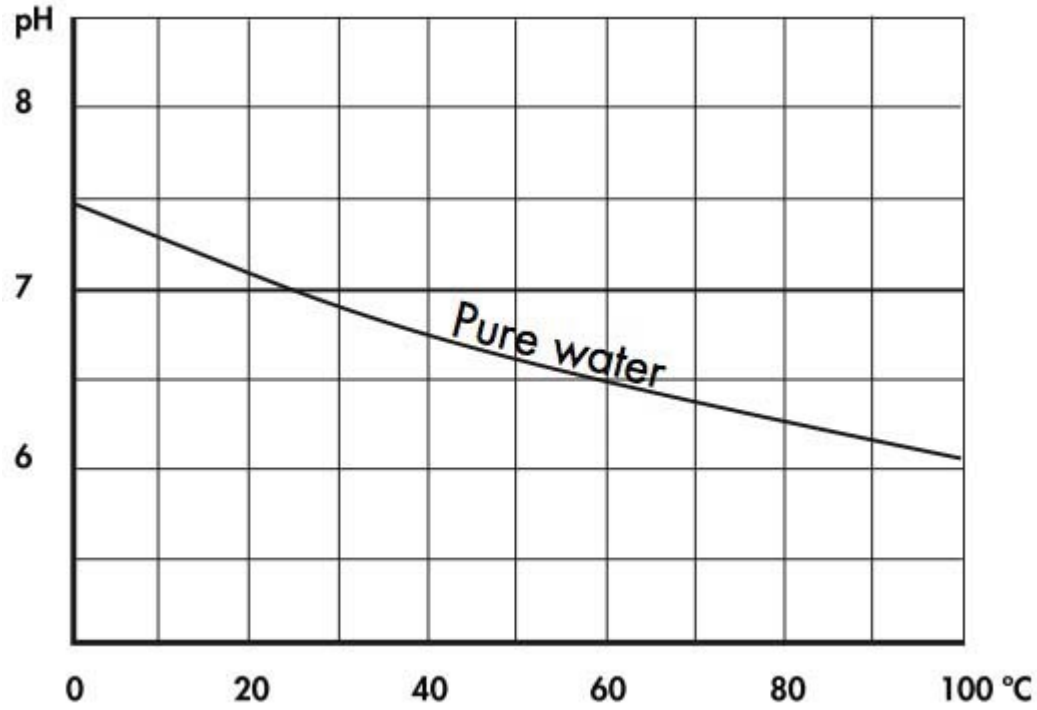


Fig. 2.2a. pH value of pure water against temperature.

Your starting point for the pH scale is pure water which is said to be neutral. Water dissociates[1] into:

$$H_20 \leftrightarrow H^+ + OH^- \quad (4)$$

Water has an equilibrium constant [2)3)]:

$$Kw = \frac{[H^+] \cdot [OH^-]}{[H_2O]} = 10^{-14} \quad (5)$$

or:

$$-\log Kw = pKw = -\log [H^+] + -\log [OH^-]$$

$$= 14 \log 10 \quad (6)$$

Pure water divides to give equal numbers of H^+ and OH^- ions and consequently, the concentrations of ions are 10^{-7} so that:

$$pH = pOH = 7$$

The pH value of pure water is 7.

This statement is incomplete, since the equilibrium constant depends on the temperature. The definition should be: The pH value of pure water is 7 @ 25°C.

Fig. 2.2a. and the table show the pH variation of pure water with temperature.

If the concentration of H^+ ions in a solution is increased (e.g. to 10^{-4}), then the solution has an acid character. In this case the pH value is lower than 7.

Some examples of common solutions with an acid character are:

$$H_2S0_4 \leftrightarrow S0_42^- + 2H^+$$

Sulphuric acid

$HCl \leftrightarrow Cl^- + H^+$
Hydrochloric acid

If the concentration of OH^- ions in a solution is increased (e.g. to 10^{-10}) then the solution is said to have a base character. In this case the pH value of the solution is a number greater than 7.

T(oC)	pKw	pH
0	14,94	7,47
18	14,22	7,11
25	14,00	7,00
50	13,22	6,61
100	12,24	6,12

Some more examples are:

$NaOH \leftrightarrow Na+ + OH^-$
Caustic soda

$NH_3^+ H_2O \leftrightarrow NH_4^+ + OH^-$
Ammonia aqueous ammonia

Note 1. See Appendix 2: Definitions
Note 2. The equilibrium constant is the ratio between the rate of decomposition and the rate of composition.
Note 3. The concentration H_2O is supposed to be 1.

pH Table

Some examples of the difference in pH value of various liquids, foods and fruit are shown in fig. 2.2b. These can be compared with the pH values of common chemical compounds dissolved in water.

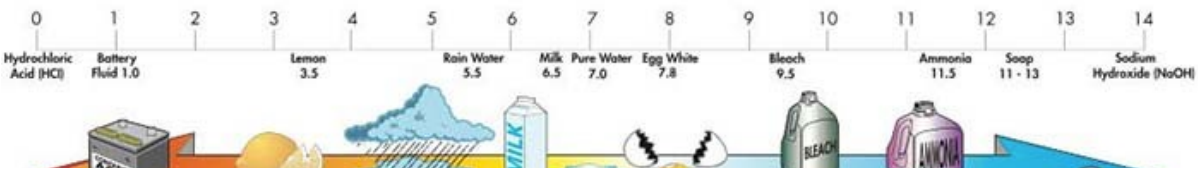




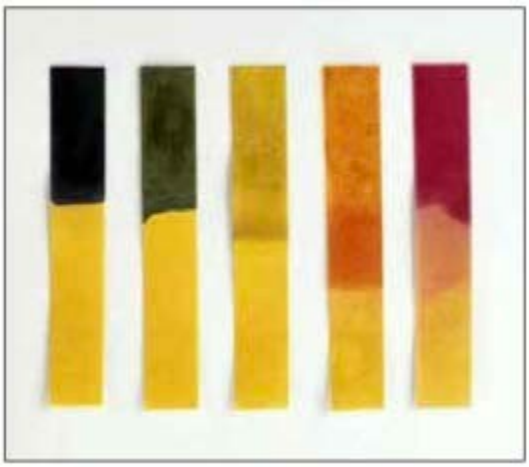
Fig. 2.2b.

2.3 Measuring the PH Scale

The pH value can be measured by different methods, e.g.:

1. Colorimetric pH measurement
2. Potentiometric pH measurement

2.3.1 Colorometric pH measurement



The principle of colorimetric determination of the pH value is based on the pH dependance of colour change.

Some examples are:

Litmus paper

When immersed in an acid medium the paper shows red, it changes to blue in a base medium. "pH paper" consists of pa- per impregnated with a suitable dye. After immersion in the liquid to be measured the colour of the wet paper can be compared with a colour disc which shows the relevant pH value for the varying shades of colour.



Some natural indicators are:

Red cabbage

Red cabbage is red in an acid medium and blue/violet in a natural medium. In an strongly basic medium the colour changes to green. Mushrooms will whiten considerably by treating with vinegar (an acid). In a base medium the mushrooms will turn brown.

2.3.2 Potentiometric pH measurement

The most often used pH sensing element is a pH sensitive glass sensor. Other pH sensors are used if a glass sensor is not acceptable (e.g. antimon sensor, ISFET). Accurate potentiometric pH will be discussed in more depth in later chapters.

2.3.3 The semiconductor sensor method (ISFET)

ISFET is a, non-glass, ion-sensitive semiconductor device (or transistor) used to measure the changes in ion concentrations within a solution. The current that passes through the transistor will change in response to the ion concentration change.

2.4 Principle of Potentiometric pH Measurement





Walther Hermann Nernst (1864-1941)

Born in Briesen, West Prussia, in 1864. He spent his early school years (Gymnasium) at Graudentz, and subsequently went to the Universities of Zurich, Berlin and Graz (Ludwig Boltzmann and Albert von Ettinghausen), studying physics and mathematics.

The principle of potentiometric pH measurement can be explained by Nernst's law.

Nernst found that a potential difference occurs between a metal object and a solution containing ions of the same metal when the object is immersed in the solution. The potential difference E, caused by the exchange of metal ions between metal and liquid, was defined by Nernst as follows:

$$E = E_{\circ} + \frac{RT}{nF} \ln [Mn^{+}] \qquad (7)$$

R = Gas constant (R=8.314J/mol.K)
F = Faraday number (F = 96493 C/ mol.)
n = Valency of the metal
$[Mn^{+}]$ = Metal ion concentration
T = Absolute temperature in Kelvin

E_o = "Normal potential"

The "normal potential" is the potential difference arising between metal and solution when this solution contains 1 mol Mn^+/litre.

Since the behavior of the gas Hydrogen has a certain degree of conformity with a metal (both have a positive ion formation), Nernst's law can also be applied to a "hydrogen electrode"[1] immersed into a solution containing hydrogen ions.

The formula can be re-written as follows:

$$E = E_o^{2)} + \frac{RT}{nF} \cdot \ln [H^+] \text{ (volt)}$$

or

$$E = \frac{RT}{F} \cdot \ln [H^+] \text{ (volt)}$$

With the constants:

$$\mathbf{E = E_o + 0{,}059 \ln [H^+] \text{ (volt)}}$$

2.5 Hydrogen Electrode, the Basic Principle

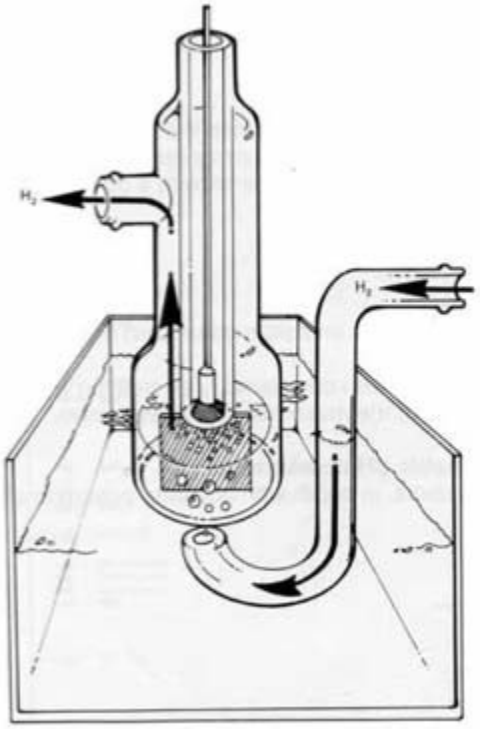


Fig. 2.5a The "hydrogen electrode"

Around 1906 Max Cremer found that some types of glass gave a potential difference of which the magnitude depends on the acid value of the liquid in which the glass was immersed.

Later, Fritz Haber and Zygmunt Klemensiewicz proved that this potential difference, within a fixed pH range, followed Nernst's law in the same way as with the so called "hydrogen electrode."

Glass can be considered as an "undercooled" electrolyte consisting of an irregular structure (SiO_2) and a number of other components which move in the interspaces. These components commonly consists of Na^+, Ca^{2+} or Li^+ -ions and give an electro-balance of the glass membrane (see figure 2.5a).

Note 1. A "hydrogen electrode" can be made by coating a layer of platinum-black on a platinum electrode and passing a flow of hydrogen gas over it. The presence of platinum-black results in the hydrogen gas being adsorbed on the electrode resulting in a so-

called "hydrogen electrode" (see: fig. 2.5a).
Note 2. By definition, the normal potential E of the metal "hydrogen" in a 1 normal H^+ solution is 0 volt at all temperatures.

When immersed in aqueous solutions, all types of pH glass have the particular property to exchange the metal-ions of the glass texture against the H^+ ions in the solution. Since the H^+ molecule are bounded to a H_20 and not free, the texture of the siliceous acid will be defound during the exchange by the bigger $H3^+$ -ion.

As a result of this reaction a so-called "gel- layer" will be developed on the surface of the glass membrane. This gel-layer is the equivalent of the metal in the Nernst's theory and is therefore essential for the functioning of the glass sensor. After one or two days the condition reaches equilibrium and the resulting gel- layer has a thickness between 10 and 40 nanometers.

This depends on several factors such as the composition of the glass and the temperature in which the sensor is immersed.

The voltage development across the glass membrane is generally explained by means of the phase limit potential theory[1)].

After reaching the equilibrium the hydrogen concentration (=activity) outside the glass and inside the gel-layer are equalized and consequently no transport of H^+ ions occurs. The voltage across the glass membrane is 0 volt.

If the concentration of hydrogen ions in the two phases differs from the concentration in the solution, a transport of hydrogen ions takes place.

The movement of the ions will affect the neutrality of the gel-layer. As a result, a voltage will be developed preventing the further

transport of H^+ ions.

The value of the voltage depends on the concentration of the hydrogen ions in the solution. Since this voltage cannot be measured directly it will be necessary to add a pH independent reference potential in the measuring circuit. This addition allows measurement of the potential differences across the glass membrane.

Note 1. For clarity, other theoretical explanations like the theories of the adsorption potential, membrane potential and statistic mechanic will not be explained.

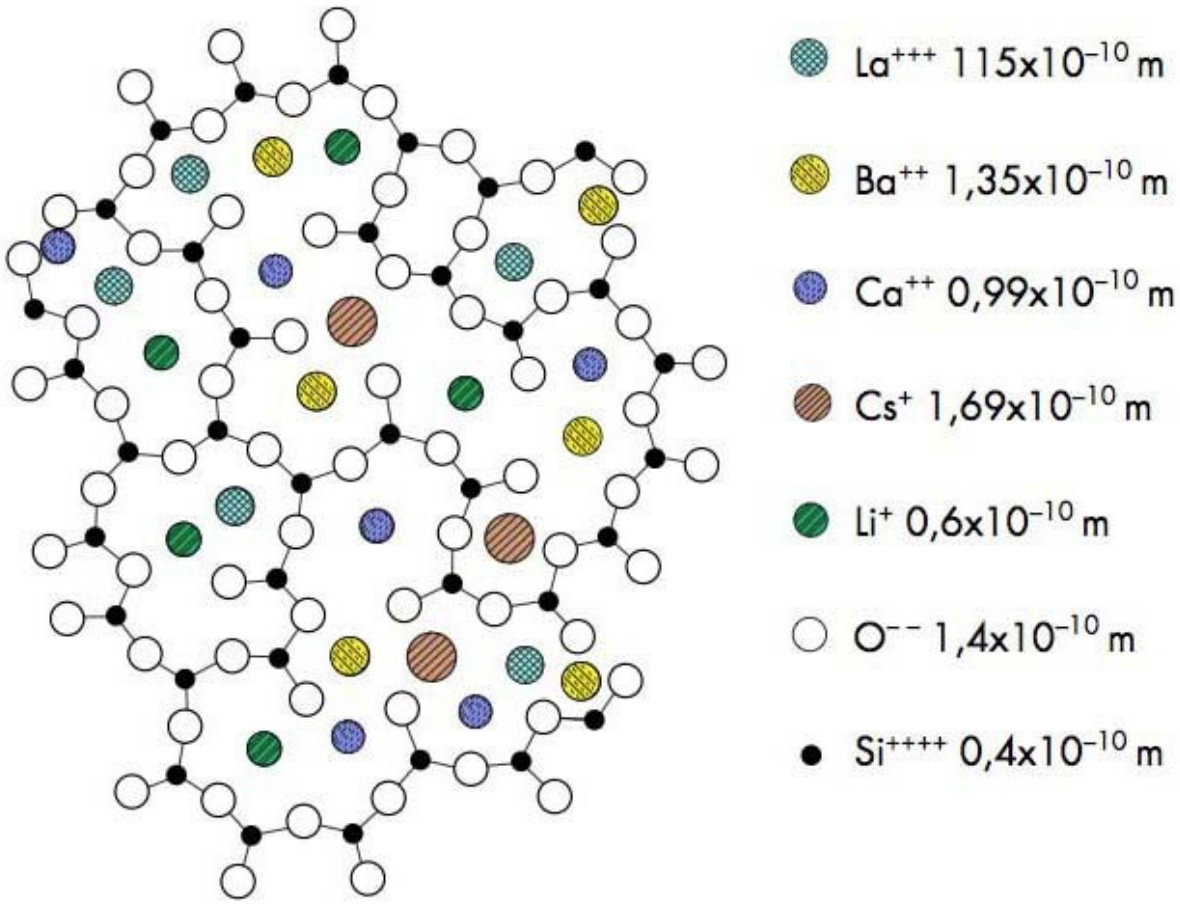


Fig. 2.5b. Texture of pH glass.

2.5.1 Composition of the Glass Electrode

Normally the glass electrode has a bulb shaped membrane of specific "pH glass" that is "welded" to the glass tube. The bulb is filled with a "buffer" solution. A reference pin is fitted and protrudes into this liquid. The complete reference system is completely separated from the other parts of the electrode and is connected to the plug of the electrode via a platinum wire welded in glass. Consequently, it is impossible for the buffer liquid to penetrate the other parts of the electrode. As the glass membrane has a high-impedance resistance, an integral metal screen which also carries a printed code denoting applications, is fitted to prevent pick-up of electrical interference. Alternatives to the standard bulb version of the "pH sensitive glass membranes" are available. Yokogawa has developed electrodes with the pH sensitive membrane designed as follows.

- **Ball Shape (Shockproof)**
  This is a universal electrode suitable for most pH applications.
- **Dome shape**
  The mechanically very strong pH membrane (thickness approx. 1 mm) is extremely suitable for measurements in aggressive media.
- **Flat shape**
  This design is used in combined sensors for application in which solids are a considerable component.

> **Note**. For a better understanding, the construction of the glass electrode and reference electrode are being shown as the single electrodes before describing the complete measuring circuit. Today these single electrodes can be combined in one pH sensor for the most applications, which be shown later.

2.5.2 Composition of the Reference Electrode

Nernst found that the combination of a metal and its insoluble salt in a salt solution produces a constant mV potential.

When such a combination (known as a reference system) is immersed directly in a process liquid, variations may occur as a result of other ions which may be present in the liquid.

Furthermore, the reference system may be poisoned by the penetration of "unwanted ions" in the salt solution. To overcome this problem an electrolyte and diaphragm is used to connect the metal/metal salt with the process liquid (see fig.2.5.2).

A constant flow of electrolyte from the electrode prevents poisoning of the electrolyte around the reference pin.

The reference system in the reference electrode is joined to the gold cable connector with a platinum pin fused in the glass.

By means of the reference electrode, it is possible to measure the pH dependent potential of the glass electrode very accurately.

Note. This potential is temperature dependent.

2.5.3 The Measuring Circuit With A Glass And A Reference Electrode

The diagrams show a pH measuring circuit using a Yokogawa pH analyser. It consists of the pH glass electrode, reference electrode, solution ground and pH analyser built as dual amplifier system. The diagrams show the potentials which effect the final potential difference (et) between the glass electrode and the reference electrode.

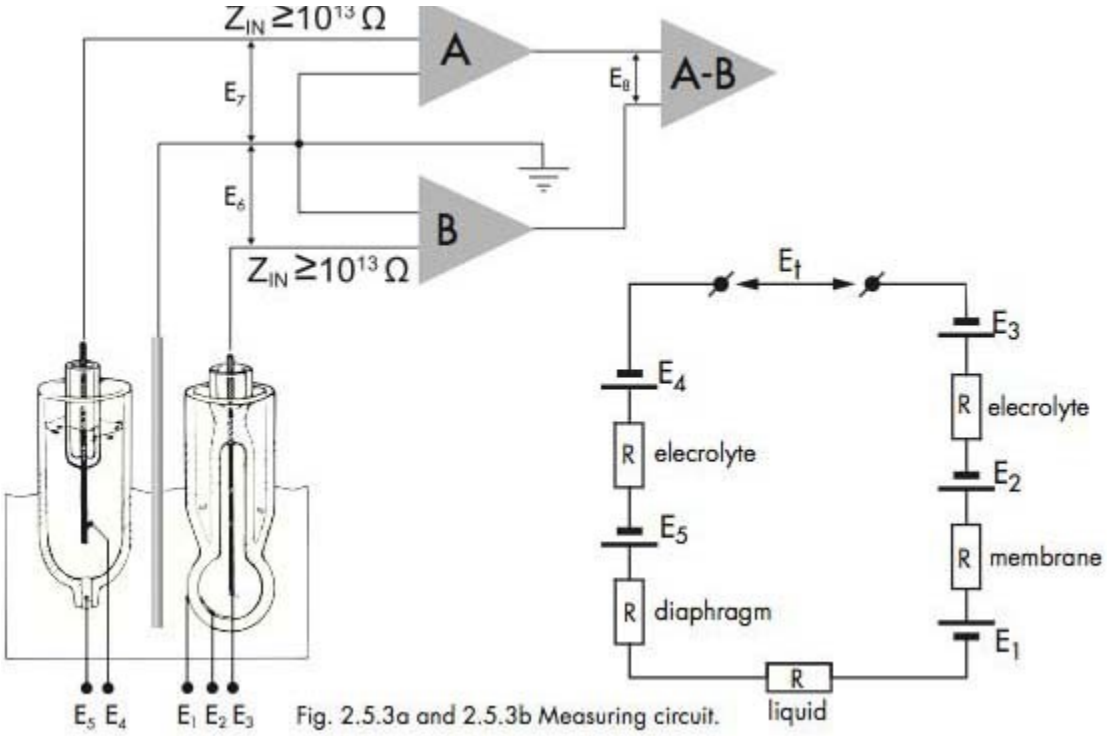


Fig. 2.5.3a and 2.5.3b Measuring circuit.

The following potentials are the most significant:

- E_1 = potential difference between the pH sensitive glass membrane and the liquid to be measured.
- E_2 = potential difference between the electrolyte in the glass electrode and the inner face of the glass membrane.
- E_3 = potential difference between the electrode pin and the electrolyte in the glass electrode.
- E_4 = potential difference between the electrolyte and the electrode pin in the reference electrode.
- E_5 = potential difference that occurs at the interface of two liquids with different concentrations, namely the electrolyte and the process liquid.
- E_6 = potential difference between pH element and solution ground at Input B of dual amplifier

- E_7 = potential difference between reference element and solution ground at Input A dual amplifier

The total sum (E_t) of these potential differences is measured by the pH- Analyser:

$$E_t = E_1 + E_2 + E_3 + E_4 + E_5 \quad (8)$$

Potential (E_6) is the potential of pH electrode 6 against solution ground:

$$E_6 = E_1 - E_2 - E_3$$

Potential (E_7) is the potential of Reference electrode against solution ground:

$$E_7 = E_4 + E_5$$

As we are only interested in the potential difference between the glass membrane and the process liquid to be measured (E_1) the remaining potentials must be compensated for so that they do not affect the true pH measurement. Re-examination of the potentials generated shows that:

If the reference systems in the glass and the reference electrode are identical and they are at the same temperature[1)], then the potentials (E_3 and E_4) generated by each are equal but opposite:

$$E_{(3)} = E_4 \Rightarrow E_{(3)} - E_4 = 0$$

The potentials E_3 and E_4 are defined as follows:

$$E = E^{\circ}_{Ne} + \frac{RT}{F} \cdot \ln\left[\frac{LmZ}{Cz^{-}}\right]$$

in which

Lmz = solubility product of sparingly soluble salt

Cz- = concentration of the salt solution

The equation (8) will then be simplified to:

$$E_t = E_1 + E_2 + E_5 \quad (9)$$

With correct selection of the electrolyte used in the reference electrode and a good flow through the liquid junction, the potential difference E_5 can be neglected, so that

$$E_t = E_1 + E_2 \quad (10)$$

$$E_1 = E^\circ + \frac{RT}{F} \cdot \ln [H^+]_{outer} \quad (11)$$

Or

$$E_1 = E^o - 0.05916 \bullet pH_{outer}$$

in the same way E_2 can be defined as:

$$E_1 = E^o - 0.05916 \bullet pH_{inner}$$

Since E_1 and E_2 in the pH measuring loop are of opposite polarities the equation becomes:

$$E_t = E_2 - E_1$$

$$E_t = 0.05916 \, (pH_{inner} - pH_{outer}) \quad (12)$$

The potential $\frac{RT}{F} \cdot pH_{inner}$ is kept constant by filling the glass electrode with an electrolyte with good buffer properties and consequently, the measured E_1 now only depends on the potential difference between the glass membrane and the process liquid. The ideal conditions described above cannot always be completely realised in practice. A small potential difference may exist when the glass and the reference electrode are both immersed in a liquid of similar properties and pH value to the electrolyte.

$$E_t = 0.05916 \, (pH_{inner} - pH_{outer}) + E_{asy} \quad (13)$$

This potential difference is called the asymmetric potential of the measuring system.

The asymmetric potential E_{asy} may be caused by:

- The liquid diffusion potential ($E_5 \neq 0$).
  The potential difference is the result of concentration differences across the flow diaphragm and is called the diffusion potential difference.
- The inner and the outer faces of the pH sensitive membrane vary because of differences in glass texture which occur during the glass blowing.
  $E_1 + E_2 \neq 0$

> **Note 1**. See chapter 2.6: "The effect of temperature".
> **Note 2**. errors resulting from easy are compensated for during calibration using buffer solutions.

2.5.4 More Information About the pH Glass Electrode

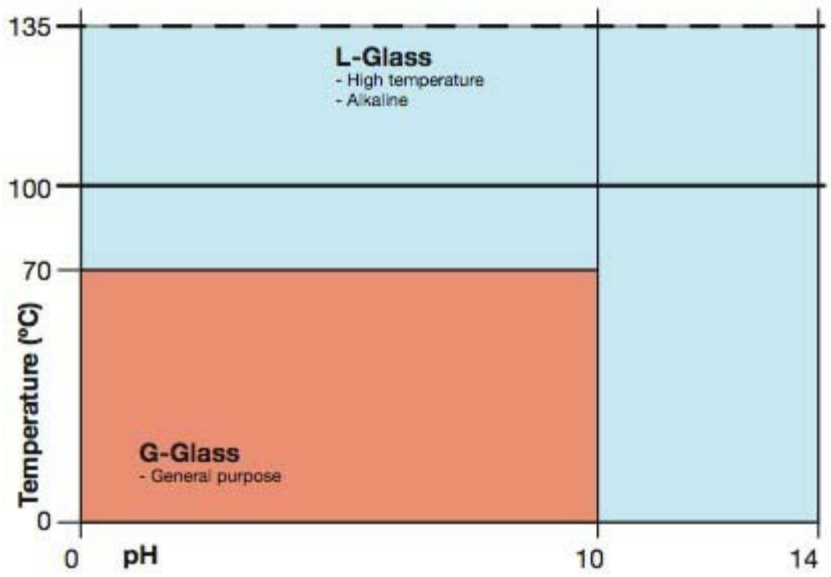


Fig. 2.5.4. Types of pH sensitive glass
and their application ranges.

The correct selection of a glass electrode for a particular application

can only be made if details of the components of the measuring loop and their significant properties are known. The following points will be considered in detail:

- selection of the glass membrane
- sensitivity of the glass electrode (mV/pH) alkaline error
- acid error
- chemical resistance of the glass membrane
- electrical resistance of the glass membrane

2.5.4.1 Selection of the glass membrane

The glass membrane reference is just as important part of the complete pH measuring loop. pH sensitive glass has the particular property that alkali metal ions present in the texture of the glass are exchanged with H^+ ions of the liquid.

To facilitate this process, it is necessary for the pH sensitive glass membrane to be "conditioned" by allowing it to absorb a film of water or gel-film. Conditioning is achieved by soaking the electrode in water for a minimum of 24 hours. The selection of the correct type of glass electrode depends on both the type and thickness of the glass membrane. Two types of glass are available, as described here:

- **"G" glass**
  This is used for the membranes of electrodes in processes where the nominal pH value varies around pH 7. Since this type of glass has a wide application range it is also termed "general purpose" glass.
- **"L" glass**

The application of "L" glass is for measurements in alkaline media with high process temperatures.

Note 1. The diagram shown in fig.2.5.4 is intended to assist with selection of the most suitable type of glass electrode in conjunction with the application range shown for each type. The range of any particular glass type also depends on the membrane thickness. Three different thicknesses are available.
Note 2. Glass electrodes manufactured by Yokogawa are "preconditioned" and may be used immediately without soaking. To form and maintain the gel-film, the sensitive glass bulb is protected with a rubber containing a small quantity of water which forms a wet pocket for the membrane.

2.5.4.2 Sensitivity of the glass electrode

The most important requirement in any electrode system for pH measurement is that the actual mV/pH ratio generated is as close as possible to the theoretical value. The potential generated by a glass electrode is given by equation:

$$E = \frac{RT}{nF} \cdot 2{,}303\ (pH_{inner} - pH_{outer})\ mV$$

At a temperature of 25°C the equation becomes:

$$E = 59{,}16 \cdot \frac{273+25}{298} \cdot (pH_{inner} - pH_{outer})\ mV$$

In the equation pH_{outer} is the pH value of the liquid at the outer face of the glass membrane and pH_{inner} is the value of the electrolyte at the inner face of the membrane.

The mV/pH ratio is called the sensitivity or slope of the electrode. The quality of the glass membrane is the most important factor in achieving correct electrode sensitivity. Reduction in sensitivity of

glass electrodes may be compensated for by adjustment of the mV/pH ratio or slope, at the analyser. The decrease in sensitivity is usually caused by fouling of the glass membrane. It is of the greatest importance that the electrode is properly cleaned before "buffering" and adjustment for sensitivity is made. If the electrode has been stored dry for a long period the sensitivity will not reach an optimal value until the electrode has been conditioned for a number of hours. When the decrease in sensitivity is caused by aging of the glass membrane the electrode can be re-activated by etching the surface of the glass membrane. This should be done by immersing the electrode for 10 seconds in a solution of vinegar (1 mol.) and potassium fluoride (1 mol.). Ratio 1:1

Note. The electrode must be cleaned carefully before and after activation. (For cleaning and re-activation see: "Direction for use" enclosed with each electrode). Frequent re-activation decreases the life of the electrode.

2.5.4.3 Alkaline of the glass electrode

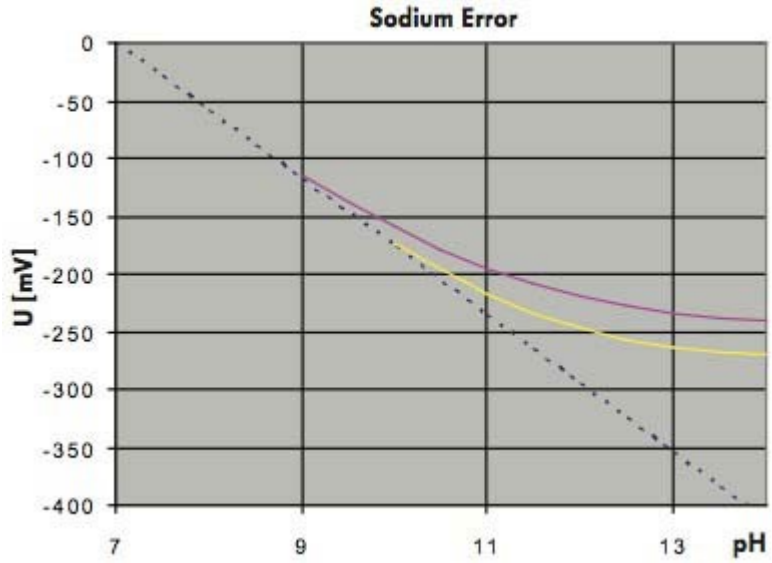


Fig. 2.5.4.3 Alkaline and acid errors of the glass electrode.

In addition to a correct mV/pH ratio it is essential that the

generation of potential difference is only influenced by the activity of the H^+ ions and not by the presence of the other monovalent cations such as Li^+, na^+, etc. A low concentration (or activity) of H^+ ions (typically a pH value of 13) and a high activity of alkaline ions may cause measuring errors of between 0.5 to 1.0 pH.

In practice, the alkaline error is often caused by sodium and consequently the term "sodium error" is also used to describe this effect.

The alkaline error can be considerable reduced by making certain additions to the pH sensitive glass which improves the selectivity of the electrode. In Fig 2.5.4.3 the alkaline error of the different types of glass at varying pH values are shown. As well as affecting the selectivity, the aforementioned additions also influence other properties, such as, chemical resistance and glass resistance.

2.5.4.4 Chemical resistance of the glass membrane

The chemical resistance of the glass membrane is greatly influenced by the process conditions. High temperatures and high salt concentrations or applications in strong alkaline liquids generally shorten the electrode life.

Additives can be included during the manufacture of the glass that make it more resistant to attack and consequently electrodes can be produced that are suitable for measurements in either strong acid or strong basic liquids. In aggressive solutions a heavy duty electrode with a thick, dome shaped, glass membrane is preferable.

2.5.4.5 Electrical resistance of the glass membrane

Since glass is a good insulator, potentiometric measurements cannot be obtained with normal glass, and constituents must be

added which will reduce the membrane resistance below 1000 MΩ to minimise the effect of electrical disturbances on the measurement.

The composition of the glass, its thickness, the surface of the glass membrane and the temperature, all affect the value of the glass resistance. Typical resistances of glass electrodes with shock-proof bulb membranes at 25°C, are as follows:

Type of glass	Membrane resistance
G-glass	50-100 MΩ
L-glass	300-500 MΩ

The thickness of the glass bulb of G glass affects the electrical resistance, as follows:

Bulb shaped (shock-proof): 50-100 MΩ
Dome shaped (heavy duty): 120-200 MΩ

2.5.4.6 The response time of the glass electrode

The response time of a glass electrode indicates the ability of an electrode to follow accurately any changes in the pH value. The response time is normally defined as the time taken to reach 63% of the value of a step change in input. Since, in practice, the response time depends on a lot of factors e.g.: the reference electrode used, the conductivity of a liquid, the temperature, the position of the electrode in the process, the process flow, the flow speed, etc. the response time quoted for a particular type is only an approximation.

Example: Glass electrode, type SM21-AG4 (shock-proof membrane).

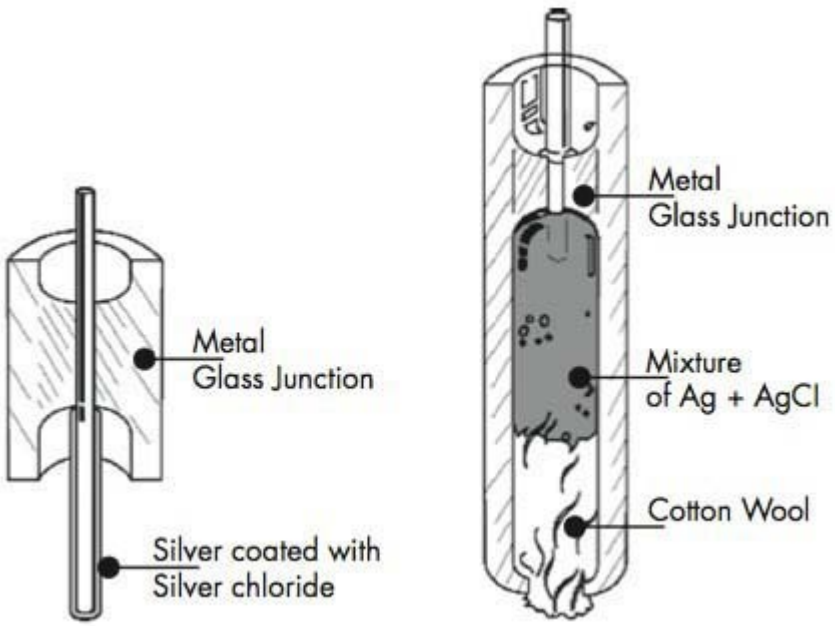
pH change: 63% of the end scale value is reached after:

1.68 to 7	5 seconds
7 to 1.68	5 seconds

2.5.5 More Information About the Reference System

2.5.5.1 General

In earlier chapters the various requirements for glass electrodes to give accurate pH measurements are described in detail. sThe accuracy of the measurement also depends on the properties of the reference electrode used. It is important therefore, to describe the different properties of reference electrodes so that a correct selection can be made.



The application range and specification are shown in table 2.5.5.2.

A good reference electrode satisfies the following requirements:

- the output voltage is determined by Nernst's law
- the output voltage is stable.

In the description below the different types of reference systems,

the flow diaphragms, and the electrolytes used in reference electrodes, are all discussed.

2.5.5.2 Reference system

Generally, the reference system used in reference electrodes are:

Silver/Silver chloride-Potassium chloride:
(Ag/ AgCl-KCl)
This reference system consists of a silver wire electrolytically coated with silver chloride. This metal - metal salt combination is dipped in a potassium chloride solution (KCl). A second type of construction for this system consists of a silver wire dipped in a paste of silver chloride, silver and potassium chloride.

This paste is sealed into a tube by means of a plug wadding soaked in KCl. This reference assembly is similarly dipped in a KCl solution.

Table 2.5.5.2. Application area and specification of various references systems.

Type of reference system	Output voltage with regards to H_2 electrode at 25°C	Application range/remarks
Silver chloride wire (AgCl) in 1 molal KCl	+223 mV ±5 mV	up to 100°C
Silver chloride paste (AgCl) in 1 molal KCl	+230 mV ±5 mV	up to 120°C

Silver-silver chloride in saturated KCl	+198 mV ±5 mV	up to 120°C

2.5.5.3 Junctions of the reference

The selection of the correct type of junction of a reference electrode depends on the process conditions under which the electrode has to function.

The following junction types are available: (see figure 2.5.6).

1. Ceramic junction.
2. Ceramic junction.
3. P.T.F.E. junction.
4. Glass sleeve capillary element.

The purpose of the junction is to maintain contact between the reference system in the electrode and the process liquid.

When selecting the correct junction, consideration has to be given to ensure that the process liquid does not penetrate into the electrode causing poisoning and a consequential unstable liquid junction potential. With the first two types of junction, listed above, the KCl solution flows slowly into the process. The flow rate is dependent on the over-pressure in the electrode and on the process temperature.

The electrolyte flow rate increases with increasing temperature. For use in very dirty liquids a glass sleeve capillary element is preferred because of its larger flow surface. The sleeve can be easily cleaned by first moving the ground ring upwards and then wiping the ground faces. non-flowing reference electrodes with a porous P.T.F.E. junction can also be used in many dirty liquid applications. The dirt resistant properties of P.T.F.E. will prevent complete fouling of the

diaphragm.

2.5.5.4 Electrolytes in the reference electrode

The electrolyte in the reference electrode must satisfy the following requirements:

- chemically inert and neutral
- no reaction with the process liquid
- having a constant activity of ions
- equitransferent i.e. the ions of the electrolyte must pass the diaphragm at equal speed
- having a low electrical resistance

The most common electrolyte used in reference electrodes are:

- 1moal KCl solution (with or without gel)
- 3.3 molal KCl solution
- saturated KCl solution.

2.5.5.5 Pressure compensated reference electrode

In processes with pressure variations, the composition of the electrolyte may change as a result of process liquid penetration into the electrode.

Any change in composition of the electrolyte may cause a measuring error or even poisoning of the reference system of the electrode.

To alleviate this problem, an electrode with an integral pressure compensation system (SR20-AC32) may be the solution. See figure 2.5.5.5.

Integral pressure compensation systems operate in a way where

the electrolyte vessel of the electrode contains bellows which are compressed in the working position. One side of the bellows is connected to the pressure via the ceramic junction and at the other side via the inner tube.

The pressure inside the bellows equals the pressure outside and only the elasticity of the bellows itself causes the over-pressure which results in a flow of electrolyte.

When the bellows are fully "expanded" the electrolyte is exhausted and refilling is required. The bellows must be compressed before refilling.

Note. The pressure compensated reference electrode can also be used in processes with pressures below atmosphere.

2.5.5.6 Reference electrode with built-in salt bridge by using double junction

In chapter 2.5.5.4 it is explained that the electrolyte in the reference electrode may not be changed by penetration of the process liquid.

Example: Mercury (Hg_2^{2+}), Copper (Cu^+), Lead (Pb_2^+) and Silver (Ag^+) ions in the process liquid will give a reaction to the KCl solution from the reference electrode.

To solve this problem the KCl solution and the process liquid must be separated using a double junction electrolyte; resulting in a reference electrode with a built-in saltbridge.

Processes containing cyanides, bromides, iodides or sulphides are a second example of selecting the KCl solution critically. Mostly, a black diaphragm indicates that the reference electrodes is used without a double junction.

The black is a deposit of silver sulphide in or directly after the flow

diaphragm. The results of such deposits can be:

- long response of the pH measuring circuit
- non-reproducible diffusion voltages and consequently drift in the indication.
- calibration is hardly possible (the formed diffusion voltage can be pH dependent).
- increased resistance of the diaphragm (slower measurement).

Note. Most biological process liquids contain sulphuric compounds.

2.5.6 Construction of the Temperature Electrode

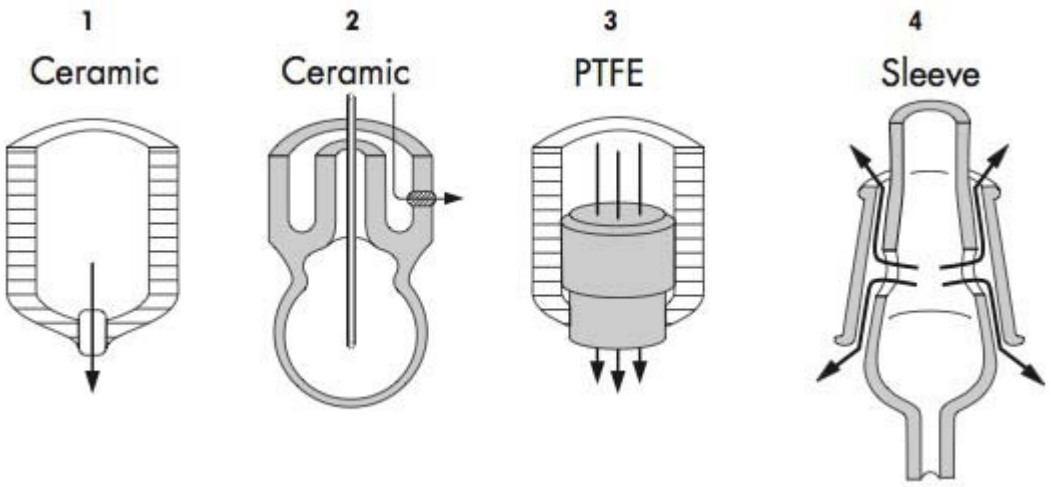


Fig. 2.5.6

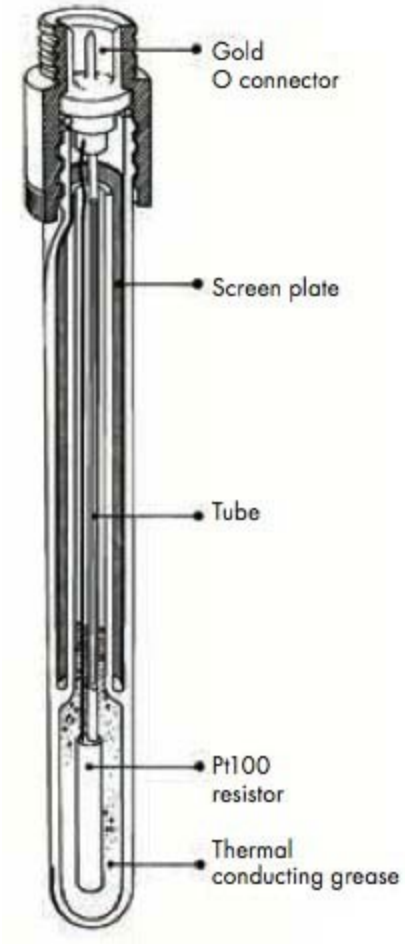


Fig. 2.5.6. The resistance thermometer

pH measurements are temperature dependent from two different effects:

1. by the variations with temperature on the contact potentials in the glass and the reference system.
2. by temperature variations of the liquid being measured.

Therefore it is necessary to include a temperature compensator in the system, whose purpose is to provide automatic compensation for the effects of temperature variations on the measuring system.

These compensators are made in the same shape as the other electrodes to enable them to be mounted in the same fittings. The temperature compensator consists of a platinum resistance element (e.g. PT100, PT1000), mounted in a glass tube. The tube

is completely filled with white silicon grease, the thermal conducting properties of which ensure fast temperature response.

As an alternative to automatic temperature compensation it is possible to manually compensate for temperature variations.

2.5.7 The Combined pH Sensors

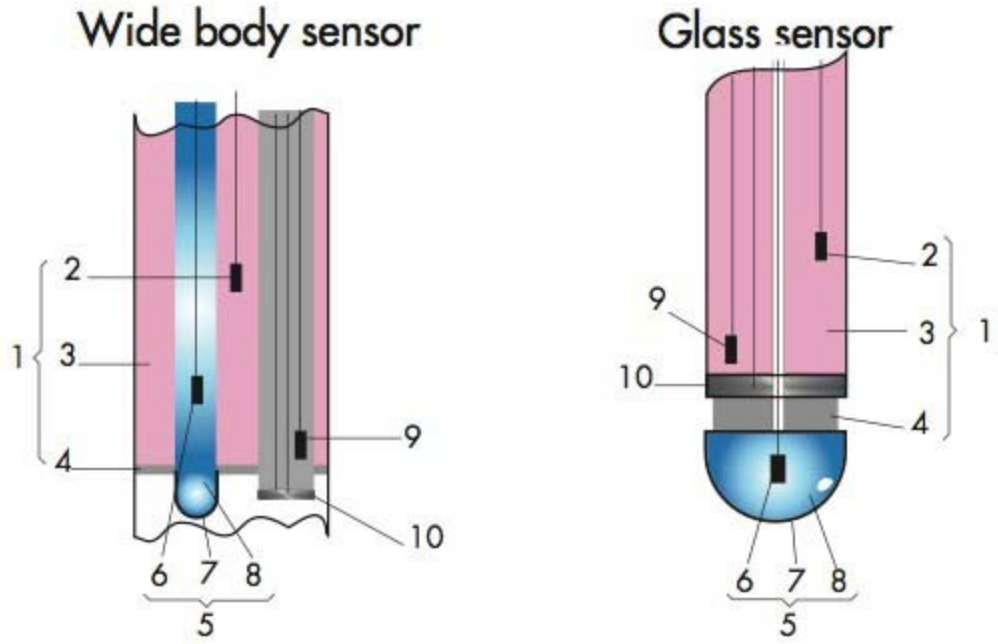
In today's business we see a tendency to use combined sensors instead of separate electrodes. In modern combined electrodes the glass-, reference-, temperature electrode and solution ground are built into one unit.

The reference systems of both the glass and the reference electrode consist of an Ag/ AgCl with same KCL solution. The operating principles are identical to those used for the individual electrodes.

The advantage of combined sensors is easy maintenance.

There has been a progression in the design of combined sensors from the ability to have just a pH and reference in a 12 mm design, to also incorporate the temperature element and the solution ground. Yokogawa has managed to fit it all electrodes into the 12 mm design with the development of the SC24V and SC25V sensors.

By incorporating the solution ground the possibilities for performance improve, along with predictive maintenance and diagnostic capabilities have improved.



1. Reference electrode
2. Reference element
3. Electrolyte
4. Juction
5. pH glass electrode
6. Internal reference element
7. Glass membrane
8. Internal buffer solution
9. Temperature element
10. Solution ground

2.6 The Effect of Temperature

2.6.1 Temperature effect on the glass and the reference electrode

The glass and the reference electrodes have a number of temperature dependent contact potentials; it is obvious then that the voltage supplied by the measuring system is temperature dependent.

This temperature dependency is shown by the factor in the Nernst equation

$$\frac{RT}{}$$

F

The voltage supplied by the measuring system is:

$$E_t = E_{asy} \times 2{,}303 \frac{RT}{F} .(pH_{inner} - pH_{outer})$$

$$E_t = E_{asy} \times 59{,}16 \left(\frac{T+273}{298}\right) (pH_{inner} - pH_{outer})$$

$\mathbf{pH_{inner}}$ is standardised at pH 7.

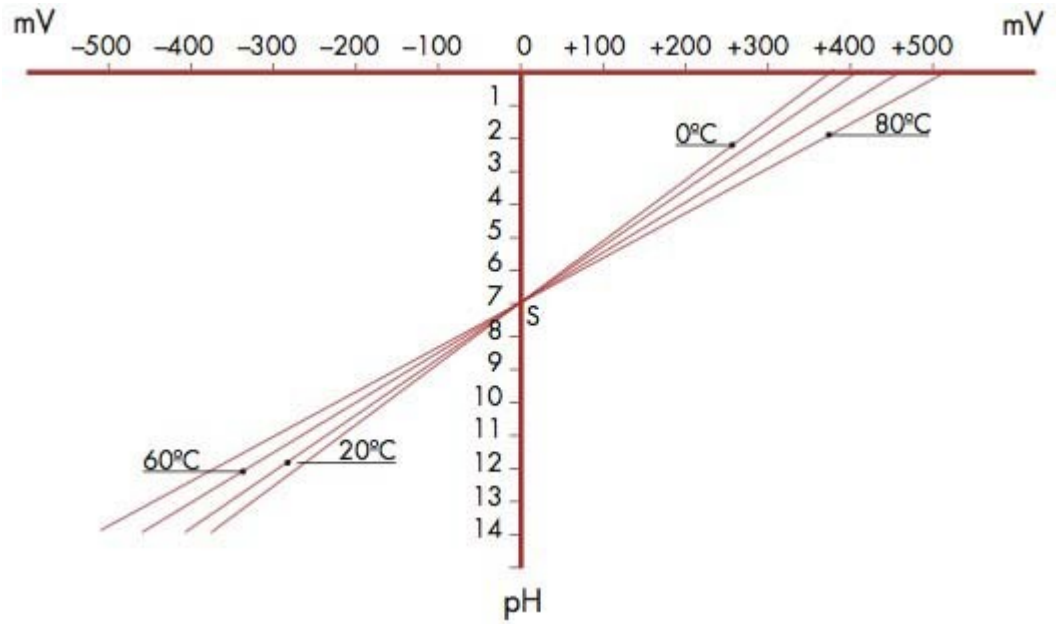


Fig. 2.6 Temperature effect on the mV/pH ratio.

T is the temperature in °C. If the glass and the reference electrodes are immersed in liquids of equal temperatures, the potential variations of similar reference systems will be equal and opposite.

$$E_3 = -E_4$$

Consequently, the system will be unaffected by temperature variations. The temperature effect on the contact potential of the junction on the reference electrode is kept to a minimum by correct selection of the junction and electrolyte. The temperature effects obtained by immersing the electrodes in different standard solutions and then by varying the temperature of these standard solutions, are shown in the graphs of fig. 2.6.

This graph shows that:

1. the mV/pH ratio increases as the temperature of the measuring system increases.
   At 25°C the mV/pH ratio is 59.16 and at 20°C this ratio is 58.16 mV/pH.
   At 80°C the mV voltage per pH unit is increased to 70.08 mV.
2. the various isothermal lines intersect at one point S (the isothermal point of intersection)
3. the intersection point is dependent on the pH of the buffer solution used in the glass electrode (this is usually pH 7).
   It is important that the isothermal lines intersect at only one point. So selection of the correct buffer solution is essential in order to obtain an accurate isothermal point of intersection S, shown in figure 2.6.

   In general, when a pH measurement is made in a process at widely fluctuating pH and temperature levels, automatic temperature compensation is necessary. To achieve this the electrode system is completed with a temperature sensing elements, packaged in a similar construction to an electrode, that compensates for slope variations of the mV/pH ratio of the electrode system.

> **Note 1**. The isothermal point of intersection of the standard electrodes of Yokogawa is at pH 7. Depending on the buffer solution used this point may, for special applications be at another value pH 3.

2.6.2 Temperature Effect on the Process Liquid

In the preceding chapter, the temperature effect on the measuring

system and its correction, has been considered. In addition there is however, a temperature effect on the chemical balance of a process itself.

It has been previously stated that the pH value of pure water at 0°C differs from the pH value at 100°C. This is caused by a change of the chemical balance.

$$H_20 \leftrightarrow H^+ + OH^-$$

at 0°C for pure water

pK =14,94 or pH =pOH =7,47

at 100°C for pure water

pK =12,24 or pH =pOH =6,12

For accurate comparison of pH values made by different techniques (e.g. by lab. measurement and industrial measurement), it is necessary to state at what temperature the measurement was made.

As the effect of temperature on any process liquid is highly dependent on its composition, it is not possible to accurately compensate for this effect automatically.

2.6.3 Temperature Effect on the Application Range of the Glass Electrode

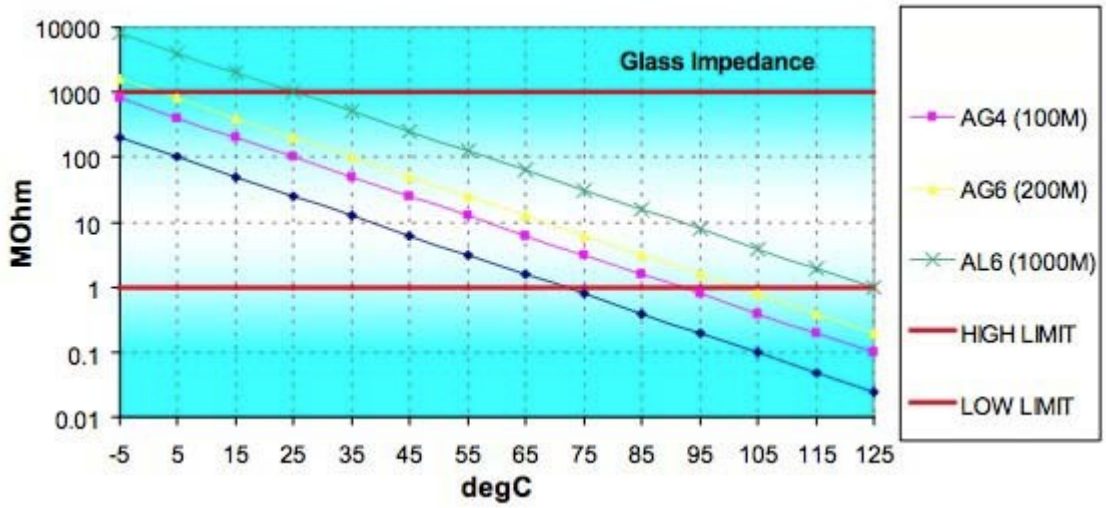


Fig. 2.6.3 Temperature effect on the glass membranes resistance.

Process temperature is a major factor in the selection of the type of glass electrode to be used for a particular application. Different reference systems are used for high or low process temperatures. Furthermore, the chemical resistance of the glass membrane is temperature dependent and correct selection is important.

A third factor is the membrane resistance of the glass electrode. This increases considerably at lower process temperatures and may increase the response time to an unacceptable level. A rough guide is that the glass membrane resistance increases by a factor 2 with every temperature fall of 10°C. Figure 2.6.3 shows the resistance of glass membranes for various species of glass.

2.6.4 NEN6411 Temperature Compensation Matrix

Using the NEN6411 norm temperature compensation can be calculated and is applicable for many applications. It's used for pH compensation in water applications using a glass-electrode. The calculation is valid for all strong acids and strong bases. The main application is in de-mineralized water and alkalised boiler feed water/condensate. The uncompensated pH value is:

$pH_{uncomp} = -\log (c[H_3O^+]) \Rightarrow c[H_3O^+] = 10^{-pHuncomp}$

where:

$$c\left[H_3O^+\right].c\left[OH^-\right] = Kw \Rightarrow c\left[OH^-\right] = \frac{Kw}{c\left[H_3O^+\right]}$$

The following relation can be derived (at reference-temperature):

$(c[H_3O^+] - d) \cdot (c[OH^-] - d) = Kw_{@\ ref\ temp}$

where:

$$Kw_t = 10^{-A} \cdots\cdots A = \frac{4471.33}{t + 273.15} + 0.017053 \cdot (t + 273.15) - 6.0846$$

Where: t, ref-temp = temperature expressed in: °C

The compensated pH value is:

$pH_{ref} = -\log (10^{-pH} - d)$

formula: 3.1.1.1

where d: = concentration change

$$d = \frac{\left(10^{-pH} + \frac{Kw_{@\ temp}}{10^{-pH}}\right) - \sqrt{\left(10^{-pH} + \frac{Kw_{@\ temp}}{10^{-pH}}\right)^2 - 4 \cdot \left(Kw_{@\ temp} - Kw_{@\ ref}\right.}}{2}$$

formula: 3.1.2.2

Where:

$Kw_{@\ temp}$ = f (Temp)
$Kw_{@\ ref\text{-}temp}$ = f (RefTemp)

2.7 Isolation Resistance

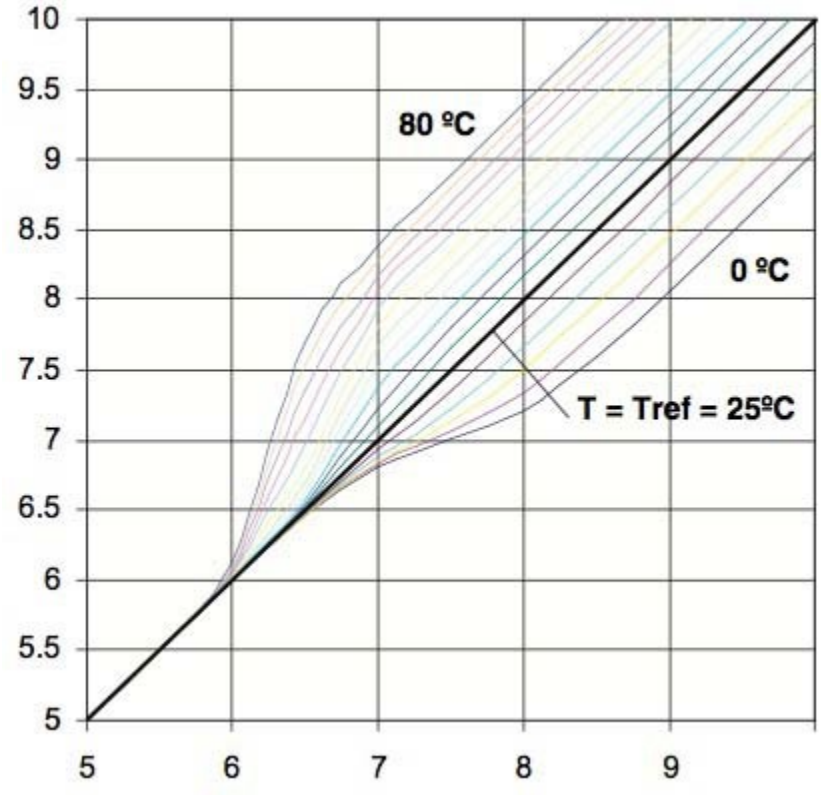


Fig. 2.6.4: NEN 6411 pH Temperature Relationship for Strong Acids and Bases

In view of the relatively high resistance of the pH sensitive glass membrane, it is necessary to use a analyser with a high input impedance. This impedance must be at least a factor of 1000 higher than the membrane resistance (the resistance of the reference electrode is much lower and can be neglected).

Insulation and screening of all cables and connections between the measuring electrode and the analyser must be of the highest order. In industrial applications the analyser should be installed as near to the electrodes as possible. At all times the connections between electrodes and analyzer should be kept dry. The insulation resistance decreases considerably when any moisture is present.

The insulation resistance of the reference electrode is less critical as its resistance with respect to the measuring liquid is much lower. Generally, an insulation resistance of $10^7\Omega$ is adequate[1)].

The resistance between reference electrode and liquid is usually between 1 and 10kW at 25°C, depending on the type of junction. At higher resistance values the sensitivity of the measuring system will be reduced and may cause an instability of measurement.

Note 1. pH measurements in low conductivity liquids with a analyser with two high input impedances for both the glass and the reference electrode require a good insulation resistance.

2.8 Buffer Solutions

Buffer solutions are needed as indispensable tool for maintaining an accurate pH measurement. Buffer solutions are used as references points for calibration and adjustment of pH measurements to compensate aging and deterioration.

Buffer solutions are mixtures of weak acids and the salt of these acids with a strong base, or mixtures of weak bases and the salt of these bases with a strong acid. Consequently, if the buffers are not accurate themselves, the calibration serves no useful purpose.

Buffers are classified in three categories. The main difference between the different types of buffers is the accuracy and buffer capacity.

Primary reference buffer

These buffers are not commercial buffer and mainly used in metrological institutes. These buffers show the lowest uncertainty in pH values, ±0.003.

Standard Buffer (secondary reference buffer)

Standard buffer solutions are used as standards for accurate measurements especially in laboratories and production of technical buffers. They are traceable to the primary standards. The

constituents of these buffers are defined by international standards like DIN19266, IeC 726 and NIST. The uncertainty is 0.002 and 0.004 pH units (at 25°C), depending on the buffer.

Technical buffer

They are commercial buffers and used mainly for calibration of industrial pH measurements. The buffer values of technical buffers are traceable to standard buffer. The DIN19267 defines standards for these solutions. The uncertainty is 0.02 a pH units (at 25°C), depending on the buffer examples of preferred buffer by Yokogawa are shown in the table below. Buffer solutions prepared from these substances conform to the recommendations of the DIN Standards Committee and the National Institute of Standards and Technology (NIST). The substances were chosen for their particular suitability as calibration standards for precision pH meters.

Standard Buffer Solutions[1)]

COMPOSITIONS	Molarity	pH at 25°C	Dilution value (pH1/2)[2)]	Buffer-capacity[3)]	Temp. coeff. dpH/dT
Potassium trihydrogen dioxalate (Tetroxalate) $KH_3(C_2O_4)2 \cdot 2H_20$	0.0496	1.679	+0,186	0,070	+0,0010
Borax $Na_2B_4O7 \cdot 10H_2O$	0.00997	9.180	+0,010	0.200	+0.0082
Potassium dihydrogen	0.02490+	6.865	+0.080	0.029	-0.0028

phosphate+ Disodium hydrogen phosphate $Na_2HPO_4 \cdot 2H_2O$ + KH_2PO_4	0.02490				
Potassium hydrogen phtalate $KHC_8H_4O_4$	0.05	4.008	+0.052	0.016	+0.0012

Note 1. N.B.S. national Bureau of Standards of the U.S.A.
Note 2,3. See Appendix 2: Definitions.

Temperature dependence

The temperature dependence of the pH of a buffer solution is generally specified in terms of measured pH values at certain discrete temperatures.

Many buffer tables are pre-programmed in Yokogawa Analyzers. So if during calibration the temperature compensator is immersed in the buffer liquid, an automatic adjustment for temperature variations will be done. Any stated pH value is only meaningful if the measuring temperature is also specified.

Be Aware

Buffers with a pH above 7 are particularly sensitive to atmospheric CO_2. Buffer showing any sign of turbidity must be discarded immediately.

For accuracy it is recommended that a buffer should not be used for more than a month after opening. Buffers should be stored in tightly sealed, preferably air-tight bottles made of polyethene or borosilicate glass. Buffers should not be returned to the bottles

once removed.

2.9 Periodic Maintenance and Calibration of pH Sensors

2.9.1 Why Is Maintenance Needed?

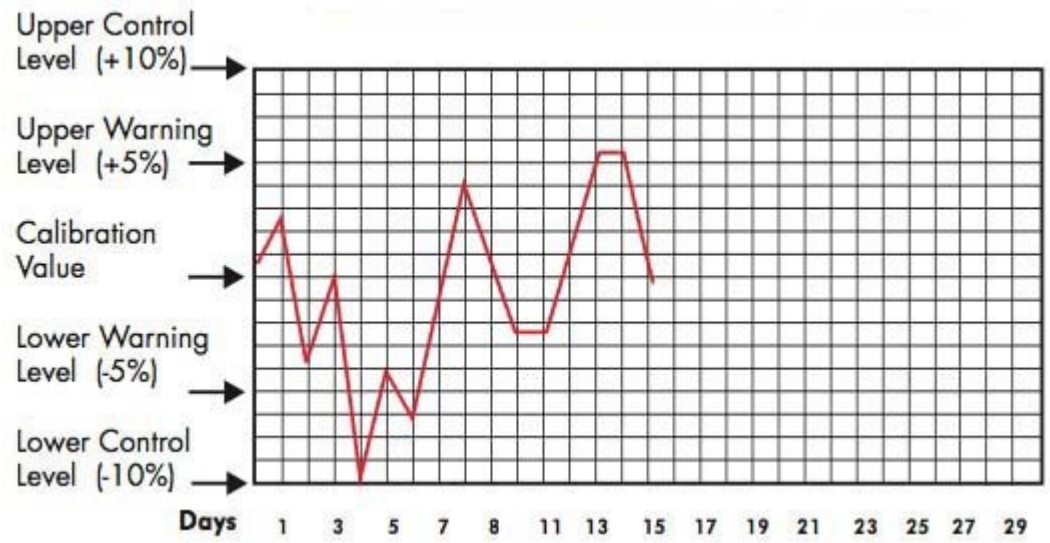


Fig. 2.9.1 Validation control chart

The selection of pH electrodes and holders (fittings) is based on the demands of the application where they will be used. The desire is to achieve an accurate, reliable measurement with a reasonable life expectancy while minimizing the required routine maintenance. When a quality pH sensor system is undamaged, clean and properly calibrated, it will provide a measurement that is accurate and reliable. This sounds simple enough, but ensuring the system is clean and calibrated will sometimes involve a significant amount of maintenance. The effect of dirty or faulty electrodes can be anything from slow response to completely erroneous measurements.

The validation control chart (Fig. 2.9.1) shows that frequency of maintenance of your measurement depends on the required accuracy. A pH measurement was checked daily in buffer solution

without adjustment. This chart shows that the reading measurement is swinging around the calibration value. To guarantee an accuracy of 5% you have to calibrate the measurement at minimum twice a week. If accuracy of 10 % is accepted you can prolong the frequency to once in two weeks.

Periodic calibration is necessary to ensure the highest measurement accuracy. Calibration adjusts for the aging of the sensors and the non-recoverable changes to the electrodes that take place.

These effects usually happen slowly therefore, calibration should not be necessary more frequently than about once a month in typical general purpose applications. If more frequent calibration is needed, it is usually because the cleaning process was not effective, the calibration was not well executed, the pH readings are temperature dependent or the wrong electrodes have been selected. If a film remains on the pH sensor after cleaning, then a measuring error can be interpreted as a need for re-calibration. Since, these changes are reversible with proper cleaning, it is a key step in the maintenance process.

> **Note**: The periodic maintenance advice that follows is intentionally gene